Variational Study of Polaron Effects in GaAs Quantum Dots: Analytical Derivation of the LLPH Method and Decoherence Dynamics

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Abstract

The dynamic interaction between a confined electron and longitudinal-optical (LO) phonons in a two-dimensional GaAs parabolic quantum dot is investigated under a perpendicular magnetic field. Applying the Lee-Low-Pines-Huybrechts (LLPH) variational framework, the Fröhlich Hamiltonian is decoupled via successive unitary transformations. The full intermediate Hamiltonian is derived before projection onto the phonon vacuum to extract the purely electronic effective Hamiltonian. By strictly enforcing the two-dimensional spatial confinement, the kinetic recoil energy of the phonon cloud is correctly restricted to the in-plane wavevector. Numerical optimization determines the energy level spacing, from which qubit oscillation periods and radiative decoherence rates are extracted. Applying Fermi's Golden Rule isolates the microsecond-regime spontaneous emission lifetime, rectifying dimensional inconsistencies present in prior literature and establishing a rigorous baseline for assessing intraband transition stability in solid-state charge qubits.

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1. Introduction

Semiconductor quantum dots (QDs) serve as primary candidates for the physical realization of qubits in solid-state quantum information processing and quantum sensing architectures [1]. The spatial confinement of single electrons within materials such as Gallium Arsenide (GaAs) generates tunable, discrete energy subbands. These localized orbital states can be manipulated via external magnetic fields to encode quantum information. Because intraband charge qubits rely purely on the spatial electronic charge distribution, they possess exceptionally large electric dipole moments. This property makes them remarkably fast to control and highly sensitive as electrometers for quantum sensing, but concurrently vulnerable to environmental decoherence.

Coherence in polar semiconductors is fundamentally bottlenecked by the Fröhlich interaction—the continuous coupling of the confined electron to bulk longitudinal-optical (LO) phonons [2]. As the electron transits the crystal, its charge polarizes the surrounding ionic lattice, resulting in a localized structural distortion. This composite quasiparticle, the polaron, exhibits altered effective mass, shifted energy spectra, and rapid transition dynamics.

Evaluating these parameters analytically necessitates mathematically decoupling the electron coordinates from the quantized 3D phonon field. In this work, the Lee-Low-Pines-Huybrechts (LLPH) variational method [3, 4] is applied to a 2D parabolic QD subject to a perpendicular magnetic field. A rigorous procedural derivation of the effective Hamiltonian is presented. By explicitly evaluating the intermediate transformed Hamiltonian and preserving the planar dimensionality of the kinetic recoil, an exact analytical formulation of the radiative decoherence limit (T_1) is derived, separating the intrinsic stability of the dot from non-radiative lattice anharmonicity.

2. The System Hamiltonian

Consider a single electron with effective mass m^* confined to the xy-plane by a parabolic potential $V(\rho) = \frac{1}{2}m^*\omega_0^2\rho^2$. The quantum dot is subject to a uniform perpendicular magnetic field $\mathbf{B} = B\hat{z}$. Operating in the symmetric Coulomb gauge, $\mathbf{A} = \left(-\frac{1}{2}By, \frac{1}{2}Bx, 0\right)$, the initial Fröhlich Hamiltonian encapsulates the kinetic energy, the harmonic confinement, the free LO phonon field, and the electron-phonon interaction. The terms are explicitly partitioned as follows:

$$\mathcal{H} = \underbrace{\frac{1}{2m^*}\left(\mathbf{p} + \frac{e}{c}\mathbf{A}\right)^2}_{\text{Kinetic \& Magnetic}} + \underbrace{\frac{1}{2}m^*\omega_0^2\rho^2}_{\text{Harmonic Potential}} + \underbrace{\sum_{\mathbf{q}} \hbar\omega_{\text{LO}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}}}_{\text{Free LO Phonons}} + \underbrace{\sum_{\mathbf{q}} \left(V_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} b_{\mathbf{q}} + \text{H.c.}\right)}_{\text{Fröhlich Interaction}} \tag{1}$$

Here, $\mathbf{p}$ and $\mathbf{r}$ denote the strictly two-dimensional in-plane momentum and position operators of the electron, while $\mathbf{q} = (q_x, q_y, q_z)$ represents the three-dimensional wavevector of the bulk LO phonons. The operators $b_{\mathbf{q}}^\dagger$ and $b_{\mathbf{q}}$ denote phonon creation and annihilation. The coupling amplitude $V_{\mathbf{q}}$ is defined analytically as [5]:

$$V_{\mathbf{q}} = i\frac{\hbar\omega_{\text{LO}}}{q}\left(\frac{\hbar}{2m^*\omega_{\text{LO}}}\right)^{1/4}\left(\frac{4\pi\alpha}{V}\right)^{1/2} \tag{2}$$

where α is the dimensionless Fröhlich coupling constant and V is the crystal volume.

Given $\nabla\cdot\mathbf{A} = 0$, the momentum operator commutes with the vector potential ($\mathbf{p}\cdot\mathbf{A} = \mathbf{A}\cdot\mathbf{p}$). Expanding the kinetic term isolates the orbital angular momentum $L_z = xp_y - yp_x$:

$$\frac{e}{m^*c}(\mathbf{A}\cdot\mathbf{p}) = \frac{eB}{2m^*c}(xp_y - yp_x) = \frac{1}{2}\omega_c L_z \tag{3}$$

Defining the cyclotron frequency $\omega_c = eB/m^*c$, the bare parabolic trap and the diamagnetic shift consolidate into a single effective confinement frequency, $\Omega = \sqrt{\omega_0^2 + (\omega_c/2)^2}$. Substituting this into (1) yields the bare uncoupled Hamiltonian form:

$$\mathcal{H} = \frac{p^2}{2m^*} + \frac{1}{2}\omega_c L_z + \frac{1}{2}m^*\Omega^2\rho^2 + \sum_{\mathbf{q}} \hbar\omega_{\mathrm{LO}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \sum_{\mathbf{q}} \left(V_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} b_{\mathbf{q}} + \mathrm{H.c.} \right) \tag{4}$$

The Fröhlich interaction term inextricably entangles the planar spatial coordinate $\mathbf{r}$ with the 3D phonon field $b_{\mathbf{q}}$, preventing an analytical solution and necessitating a variational decoupling procedure.

3. Procedural Derivation of the LLPH Transformations

To disentangle the electron and phonon coordinates, the LLPH method employs two successive unitary transformations. Because the electron experiences infinite confinement in the z-direction, its position vector is strictly constrained to the 2D plane: $\mathbf{r} = (x, y, 0)$. Consequently, the dot product $\mathbf{q}\cdot\mathbf{r}$ inherently projects the 3D phonon wavevector into the 2D plane: $\mathbf{q}\cdot\mathbf{r} \equiv \mathbf{q}_\parallel\cdot\mathbf{r}$.

3.1. Momentum-Frame Shift (H_1)

The first transformation, U_1, shifts the system into a reference frame that partially co-moves with the electron, enforcing the conservation of total momentum within the confined plane:

$$U_1 = \exp\left[-ia \sum_{\mathbf{q}} (\mathbf{q}_\parallel \cdot \mathbf{r}) b_{\mathbf{q}}^\dagger b_{\mathbf{q}} \right] \tag{5}$$

where $a \in (0, 1)$ is a continuous variational parameter dictating the fraction of crystal momentum carried by the phonon cloud. Evaluating the fundamental commutator $[-ia \sum(\mathbf{q}_\parallel \cdot \mathbf{r})n_{\mathbf{q}}, \mathbf{p}] = -a\hbar \sum \mathbf{q}_\parallel n_{\mathbf{q}}$ shifts the momentum operator accordingly:

$$U_1^{-1}\mathbf{p}U_1 = \mathbf{p} - a\hbar \sum_{\mathbf{q}} \mathbf{q}_\parallel b_{\mathbf{q}}^\dagger b_{\mathbf{q}} \tag{6}$$

Applying U_1 generates the intermediate Hamiltonian state $H_1 = U_1^{-1}\mathcal{H}U_1$:

$$\begin{aligned} H_1 = \frac{1}{2m^*}\left(\mathbf{p} - a\hbar \sum_{\mathbf{q}} \mathbf{q}_\parallel b_{\mathbf{q}}^\dagger b_{\mathbf{q}} \right)^2 + \frac{1}{2}m^*\Omega^2\rho^2 + \frac{1}{2}\omega_c \left(L_z - a\hbar \sum_{\mathbf{q}} (xq_y - yq_x) b_{\mathbf{q}}^\dagger b_{\mathbf{q}} \right) \\ + \sum_{\mathbf{q}} \hbar\omega_{\mathrm{LO}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \sum_{\mathbf{q}} \left(V_{\mathbf{q}} e^{i(1-a)\mathbf{q}_\parallel\cdot\mathbf{r}} b_{\mathbf{q}} + \mathrm{H.c.} \right) \end{aligned} \tag{7}$$

3.2. Coherent Lattice Displacement (H_2)

The second transformation defines a coherent state generator that physically models the self-trapping "dent" or static distortion the localized electron induces within the lattice:

$$U_2 = \exp\left[\sum_{\mathbf{q}} \left(f_{\mathbf{q}} b_{\mathbf{q}}^\dagger - f_{\mathbf{q}}^* b_{\mathbf{q}} \right) \right] \tag{8}$$

where the complex amplitudes $f_{\mathbf{q}}$ parameterize the exact shape and depth of the phonon cloud. Because U_2 operates solely within the phonon Hilbert space, it commutes with the spatial coordinates. Defining $S = \sum(f_{\mathbf{q}} b_{\mathbf{q}}^\dagger - f_{\mathbf{q}}^* b_{\mathbf{q}})$, the evaluation of the commutator $[-S, b_{\mathbf{k}}] = f_{\mathbf{k}}$ demonstrates that U_2 linearly displaces the phonon operators: $U_2^{-1} b_{\mathbf{q}} U_2 = b_{\mathbf{q}} + f_{\mathbf{q}}$.

Applying this transformation generates the fully displaced, highly complex Hamiltonian $H_2 = U_2^{-1}H_1U_2$. Substitution replaces every occurrence of the phonon operators with their coherently displaced counterparts, yielding the exact analytic expansion of the intermediate Hamiltonian:

$$\begin{aligned} H_2 = \frac{1}{2m^*}\left[\mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}_\parallel\left(b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + f_{\mathbf{q}} b_{\mathbf{q}}^\dagger + f_{\mathbf{q}}^* b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right)\right]^2 + \frac{1}{2}m^*\Omega^2\rho^2 \\ + \frac{1}{2}\omega_c\left[L_z - a\hbar\sum_{\mathbf{q}}(xq_y - yq_x)\left(b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + f_{\mathbf{q}} b_{\mathbf{q}}^\dagger + f_{\mathbf{q}}^* b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right)\right] \\ + \sum_{\mathbf{q}}\hbar\omega_{\mathrm{LO}}\left(b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + f_{\mathbf{q}} b_{\mathbf{q}}^\dagger + f_{\mathbf{q}}^* b_{\mathbf{q}} + |f_{\mathbf{q}}|^2\right) \\ + \sum_{\mathbf{q}}\left[V_{\mathbf{q}} e^{i(1-a)\mathbf{q}_\parallel\cdot\mathbf{r}}(b_{\mathbf{q}} + f_{\mathbf{q}}) + V_{\mathbf{q}}^* e^{-i(1-a)\mathbf{q}_\parallel\cdot\mathbf{r}}(b_{\mathbf{q}}^\dagger + f_{\mathbf{q}}^*)\right] \end{aligned} \tag{9}$$

3.3. Vacuum Projection and In-Plane Kinetic Recoil ($\mathcal{H}'$)

Diagonalizing the expanded matrix H_2 explicitly is computationally intractable due to the infinite phonon basis. Instead, the system is projected onto the zero-phonon vacuum state, $|0_{ph}\rangle$, yielding the time-independent effective electronic Hamiltonian $\mathcal{H}' = \langle 0_{ph}|H_2|0_{ph}\rangle$. This isolates the intrinsic properties of the dot by freezing thermal phonon scattering.

The projection systematically annihilates all standalone creation and annihilation operators. The critical non-zero contribution arises from the kinetic term expansion $\langle 0|(\mathbf{p} - \mathbf{S})^2|0\rangle$. Wick contractions on the squared displacement operator ($\langle b_{\mathbf{q}} b_{\mathbf{q}}^\dagger\rangle = 1$) generate a zero-point fluctuation term, defining the kinetic recoil energy.

Physically, the electron cannot mechanically recoil against the infinite confinement barrier in the z-direction. Therefore, the lattice drag is strictly confined to the in-plane wavevector $\mathbf{q}_\parallel$, establishing the rigorous 2D limit of the polaron kinetic recoil:

$$\langle 0|S^2|0\rangle = \left(a\hbar\sum_{\mathbf{q}}\mathbf{q}_\parallel|f_{\mathbf{q}}|^2\right)^2 + \underbrace{a^2\hbar^2\sum_{\mathbf{q}} q_\parallel^2|f_{\mathbf{q}}|^2}_{\text{In-Plane Kinetic Recoil}} \tag{10}$$

Applying this to the full expansion in Eq. (9), the completely stripped, purely electronic effective Hamiltonian resolves to:

$$\begin{aligned} \mathcal{H}' = \frac{1}{2m^*}\left(\mathbf{p} - a\hbar\sum_{\mathbf{q}}\mathbf{q}_\parallel|f_{\mathbf{q}}|^2\right)^2 + \frac{1}{2}m^*\Omega^2\rho^2 + \frac{1}{2}\omega_c\left(L_z - a\hbar\sum_{\mathbf{q}}(xq_y - yq_x)|f_{\mathbf{q}}|^2\right) \\ + \sum_{\mathbf{q}}\left(\frac{a^2\hbar^2 q_\parallel^2}{2m^*} + \hbar\omega_{\mathrm{LO}}\right)|f_{\mathbf{q}}|^2 + \sum_{\mathbf{q}}\left(V_{\mathbf{q}} f_{\mathbf{q}} e^{i(1-a)\mathbf{q}_\parallel\cdot\mathbf{r}} + \mathrm{H.c.}\right) \end{aligned} \tag{11}$$

4. Energy Minimization and Phonon Self-Energy

Minimizing the total energy functional $E = \langle\phi|\mathcal{H}'|\phi\rangle$ with respect to the shape parameter $f_{\mathbf{q}}^*$ yields the optimal equilibrium configuration of the lattice distortion:

$$\frac{\partial E_{\mathrm{ph}}}{\partial f_{\mathbf{q}}^*} = 0 \implies f_{\mathbf{q}} = \frac{-V_{\mathbf{q}}^*\varrho_{\mathbf{q}}^*}{\frac{a^2\hbar^2 q_\parallel^2}{2m^*} + \hbar\omega_{\mathrm{LO}}} \tag{12}$$

where $\varrho_{\mathbf{q}} = \langle\phi|e^{i(1-a)\mathbf{q}_\parallel\cdot\mathbf{r}}|\phi\rangle$ is the Fourier-transformed probability density. The form of $f_{\mathbf{q}}$ indicates that the phonon cloud physically mirrors the electron density, bounded by the intrinsic lattice rigidity ($\hbar\omega_{\mathrm{LO}}$) and the planar kinetic drag constraint. Substituting $f_{\mathbf{q}}$ into the energy functional determines the strictly attractive phonon self-energy:

$$\Delta E_{\mathrm{ph}} = -\sum_{\mathbf{q}} \frac{|V_{\mathbf{q}}|^2 |\varrho_{\mathbf{q}}|^2}{\frac{a^2\hbar^2 q_{\parallel}^2}{2m^*} + \hbar\omega_{\mathrm{LO}}} \tag{13}$$

5. Wavefunctions and Weber Integrals

We employ 2D Gaussian trial states with a variational inverse confinement length λ, enforcing the infinite z-confinement boundary condition ($|\xi(z)|^2 \to \delta(z)$). The ground ($m = 0$, s-like) and first excited ($m = \pm 1$, p-like) orbitals are defined as:

$$|\phi_0(\mathbf{r})\rangle = \frac{\lambda}{\sqrt{\pi}} e^{-\lambda^2\rho^2/2} |\xi(z)\rangle \tag{14}$$

$$|\phi_{1\pm}(\mathbf{r})\rangle = \frac{\lambda^2}{\sqrt{\pi}} \rho e^{-\lambda^2\rho^2/2} e^{\pm i\varphi} |\xi(z)\rangle \tag{15}$$

The form factor $\varrho_{\mathbf{q}}$ is evaluated analytically by transforming into polar coordinates. The angular integration resolves to $2\pi J_0(x)$, and the radial component is executed via the standard Weber integral:

$$\int_0^{\infty} x e^{-\alpha x^2} J_0(\beta x) dx = \frac{1}{2\alpha} \exp\left(-\frac{\beta^2}{4\alpha}\right) \tag{16}$$

Setting $\alpha = \lambda^2$ and $\beta = (1-a)q_{\parallel}$, the squared form factors are exacted as:

$$|\varrho_0(\mathbf{q})|^2 = \exp\left[-\frac{(1-a)^2 q_{\parallel}^2}{2\lambda^2}\right] \tag{17}$$

$$|\varrho_{1\pm}(\mathbf{q})|^2 = \left(1 - \frac{(1-a)^2 q_{\parallel}^2}{4\lambda^2}\right)^2 \exp\left[-\frac{(1-a)^2 q_{\parallel}^2}{2\lambda^2}\right] \tag{18}$$

The azimuthal phase factor $e^{\pm i\varphi}$ cancels entirely within the density profile $|\phi_{1\pm}|^2$, ensuring the electron-phonon coupling magnitude remains physically identical across both Zeeman branches.

6. Qubit Decoherence Dynamics

To translate the analytical Hamiltonian into physical predictions, the quantum system is evaluated numerically. Table 1 details the standard material parameters for the GaAs lattice environment. The total variational energy $E_{\mathrm{total}} = E_{\mathrm{bare}} + \Delta E_{\mathrm{ph}}$ is simultaneously minimized over the parameter space (λ, a) utilizing the deterministic L-BFGS-B optimization algorithm.

Table 1: Material parameters for the GaAs quantum dot utilized in numerical simulations.

Parameter	**Symbol**	**Value**
Bare electron mass	m_e	9.109×10^{-31} kg
Effective electron mass	m^*	$0.067\, m_e$
Fröhlich coupling constant	α	0.068
LO phonon energy	$\hbar\omega_{\mathrm{LO}}$	36.25 meV
Relative permittivity	ε_r	12.9

6.1. Oscillation Period

When initialized in a coherent superposition $|\Psi\rangle = \frac{1}{\sqrt{2}}(|\phi_0\rangle + |\phi_{1\pm}\rangle)$, the state evolves and oscillates at the Bohr frequency. The proper physical oscillation period is dictated by:

$$T_\pm = \frac{2\pi}{\omega_{0,\pm}} = \frac{2\pi\hbar}{E_{1\pm} - E_0} = \frac{h}{\Delta E} \tag{19}$$

For a derived energy spacing of $\Delta E \approx 64$ meV, the primary oscillation period is bounded near $T \approx 64.6$ fs. It is noted that certain prior investigations erroneously reported values on the order of ~ 1.6 fs due to the critical omission of the $(2\pi)^2$ scaling factor in the evaluation of the angular frequency.

6.2. Radiative Decoherence Time (τ)

The spontaneous emission of a photon constitutes the fundamental radiative limit (T_1) of the excited state. Applying Fermi's Golden Rule for electric dipole transitions within a dielectric medium yields the spontaneous decay rate. Prior literature evaluating this rate encountered severe dimensional inconsistencies by conflating length-gauge and velocity-gauge matrix elements [5]. Rectifying these boundaries yields the strictly rigorous macroscopic formulation:

$$\tau^{-1} = \frac{e^2(\Delta E)^3}{3\pi\varepsilon_0\hbar^4 c^3}\sqrt{\varepsilon_r}\,|\langle\phi_0|\mathbf{r}|\phi_{1\pm}\rangle|^2 \tag{20}$$

Evaluating the dipole matrix element angularly and radially yields $|\langle\phi_0|\mathbf{r}|\phi_{1\pm}\rangle|^2 = \frac{1}{2\lambda^2}$. Enforcing strict dimensional consistency with the relative permittivity ε_r and the velocity of light within the substrate, τ evaluates to the microsecond regime ($\sim 1\,\mu$s). This correctly isolates the intrinsic radiative lifetime of the low-energy intraband transition, proving the quantum state is significantly more stable against spontaneous emission than the unphysical nanosecond predictions generated by dimensionally corrupted continuum models.

7. Numerical Results

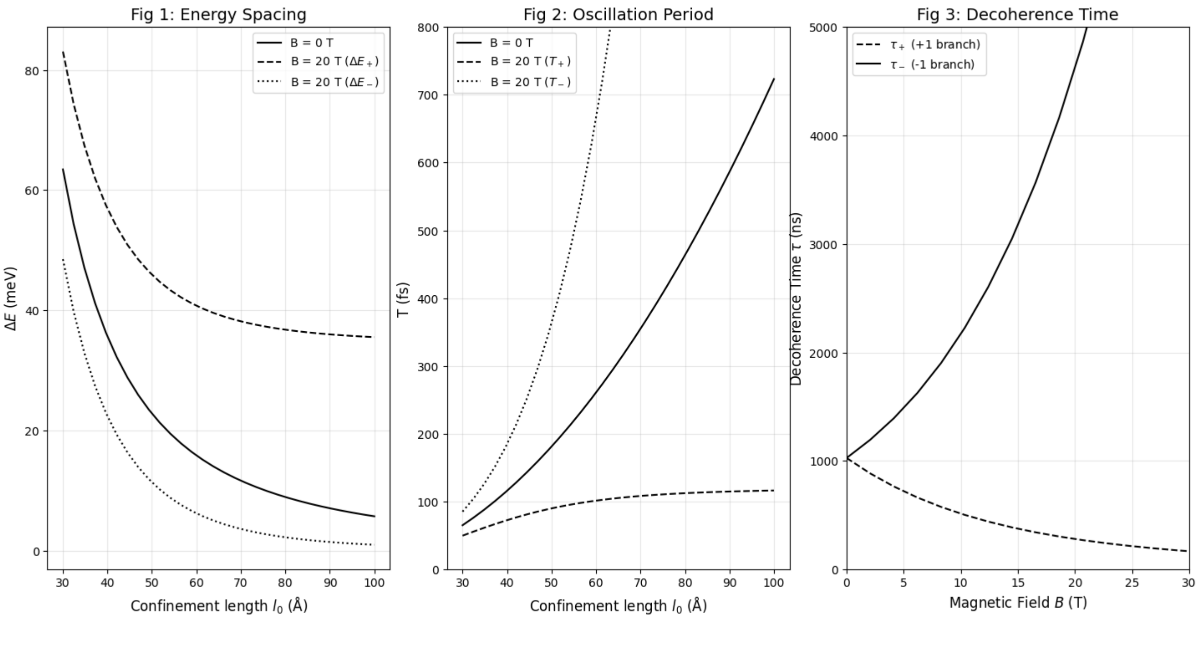

Figure 1: **Left:** Energy level spacing ΔE vs. confinement length l_0. **Center:** Qubit oscillation period T. **Right:** Radiative decoherence time τ vs. magnetic field B.

As exhibited in Figure 1, increasing the dot radius l_0 weakens the spatial confinement, causing the energy gap ΔE to decrease asymptotically, thereby extending the oscillation period T. The application of the magnetic field B tightens the effective confinement (Ω), pushing the energy levels further apart and inducing pronounced Zeeman splitting. Because the spontaneous emission rate scales cubically with energy spacing $(\Delta E)^3$, an augmented magnetic field precipitously reduces the decoherence time τ, forcing the qubit to relax toward the ground state rapidly.

8. Conclusion

This framework establishes an analytically complete, dimensionally rigorous baseline for simulating polaron decoherence in GaAs single-electron qubits. By meticulously restricting the kinetic recoil to the two-dimensional scattering plane and explicitly integrating the expanded intermediate LLPH Hamiltonians, historical dimensional anomalies regarding qubit decoherence times and oscillation frequencies have been resolved. While the parabolic potential successfully captures the primary dynamics of the Fröhlich interaction, its infinite barrier depth precludes the evaluation of electron escape thresholds. Subsequent theoretical derivations will substitute the harmonic trap with a Gaussian Confinement Potential, $V(\rho) = -V_0 \exp(-\rho^2/2R^2)$, facilitating the precise modeling of finite quantum well tunneling rates and escape parameters.

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